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25 May 1995
SUPERSEDING
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27 June 1989

MILITARY SPECIFICATION

MODULE, FIBER OPTIC, TRANSMITTER, DIGITAL, 160 MBD, ASSOCIATED DETAIL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-24791.

1. SCOPE

1.1 Scope. This specification, in conjunction with MIL-M-24791, describes a digital fiber optic transmitter for use in SAFENET LAN installations. It may also be used in other applications where the modulation rate does not exceed 160 MBd (megabaud).

1.2 Classification. This specification applies to two types of transmitter.

Type LP - for use in SAFENET class one and class two systems
Type HP - for use in SAFENET class three systems

All requirements apply to both types except where indicated otherwise.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-49291/6 - Fiber Optical, 62.5/125 Micrometers, Radiation Harden (Metric) Associated Detail Specification For

MIL-H-38534 - Hybrid Microcircuits, General Specification For

MIL-M-24791 - Module, Fiber Optic, Transmitter or Receiver, Digital, General Requirements For

STANDARDS

MILITARY

MIL-STD-883 - Test Methods And Procedures For Microelectronics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command, SEA 05Q42, Washington, DC 20362-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A PSC 6025

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS

IEEE STD 194-1977 - Pulse Terms And Definitions

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION/INTERNATIONAL ELECTROTECHNICAL COMMISSION (ISO/IEC)

ISO/IEC 9314-3 - Information Processing System - Fiber Distributed Data Interface (FDDI) - Part 3: Physical Layer Medium Dependent (PMD)

(Applications for copies should be addressed to the Global Engineering Documents, 2805 McGaw Avenue, Irvine, California 92714.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Concurrent application of requirements. Unless otherwise specified, all requirements herein and of MIL-M-24791 apply concurrently.

3.2 Functional.

3.2.1 Signal ports. Signal inputs and output shall be provided, as specified by the subparagraphs to this paragraph.

3.2.1.1 Inputs. All inputs shall be electrical interfaces in accordance with 3.2.2. Required inputs are identified in the subparagraphs to this paragraph.

3.2.1.1.1 Data. Two inputs, +DATA and -DATA, in accordance with 3.2.5.1 and 3.2.6.

3.2.1.1.2 Disable control. A transmit disable input in accordance with 3.2.5.2

3.2.1.2 Output. A single output providing an optical signal in accordance with 3.2.3, 3.2.4, 3.2.5, and 3.2.6.

3.2.2 Electrical interface. Electrical inputs shall conform to the definition for emitter coupled logic inputs in MIL-M-24791, Appendix A.

3.2.3 Optical carrier.

3.2.3.1 Reference fiber. The reference fiber is that described by MIL-F-49291/6.

3.2.3.2 Optical spectrum. Central wavelength and spectral width of the modulated optical carrier shall be as specified in table I.

TABLE I. Spectrum.

Parameter	Requirement	Notes
Central wavelength, λ_c	1260 nm $< \lambda_c <$ 1380 nm	1/
Spectral width, $\Delta\lambda$	40 nm $\leq \Delta\lambda \leq$ 210 nm, for $\lambda_c >$ 1315 nm 40 nm $< \Delta\lambda <$ [210-1.8(1315-λ_c)], for $\lambda_c <$ 1315 nm	1/

1/ See paragraph 6.3 of MIL-M-24791.

3.2.3.3 Power and modulation extinction ratio. Power and modulation extinction ratio shall be specified in table II when modulated as described by 3.2.4. See paragraph 3.2.5.2 for transmit disable requirements.

TABLE II. Power.

PARAMETERS	TYPE	TRANSMIT DISABLE INPUT	MINIMUM	MAXIMUM	UNITS
POWER	LP	LOW	-20	-14	dBm
		HIGH		-49	dBm
	HP	LOW	-16	-9	dBm
		HIGH		-49	dBm
EXTINCTION RATIO	BOTH	LOW	10		

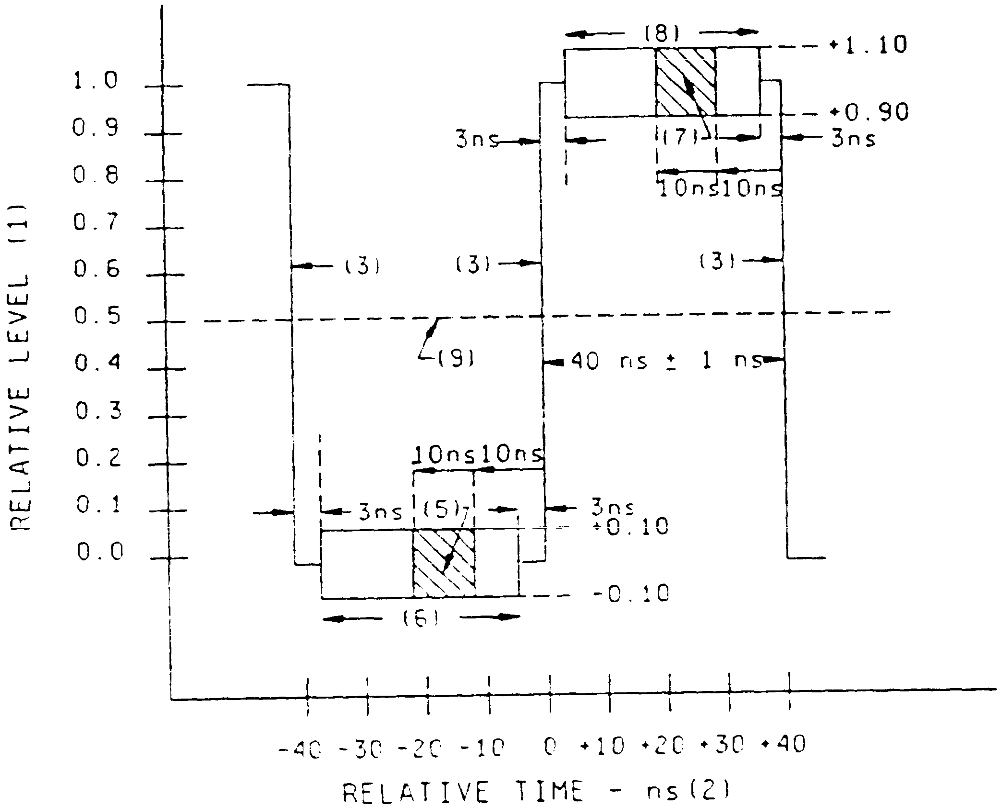
3.2.4 Envelope waveform. The envelope waveform shall lie within the bounds specified by templates defined by figures 1, 2 and 3 and by 3.2.4.1 and 3.2.4.2 when the DATA input pattern is a 12.50 MHz squarewave. Functional requirements apply for all DATA input patterns having characteristics within the ranges specified by 3.2.4.3.

3.2.4.1 Base and top (see 6.3). Base and top are the average of the waveform within the L(LR) and L(HR) regions, respectively, indicated in figure 1.

3.2.4.2 Transition waveform (see 6.3). Transition waveforms are those portions of the envelope waveform within the interval extending from +3 ns to -3 ns with respect to the start or stop line. Transition waveforms shall be as specified in the subparagraphs to this paragraph.

3.2.4.2.1 Positive. The template of figure 2 specifies the bounds for transitions from base to top.

3.2.4.2.2 Negative. The template of figure 3 specifies the bounds for transitions from top to base



1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
RELATIVE LEVEL (1)
(8)
+1.10
+0.90
3ns
(7)
3ns
10ns
10ns
(3)
(3)
(3)
(9)
40 ns ± 1 ns
10ns
10ns
3ns
(5)
3ns
+0.10
-0.10
(6)
-40
-30
-20
-10
0
+10
+20
+30
+40
RELATIVE TIME - ns (2)

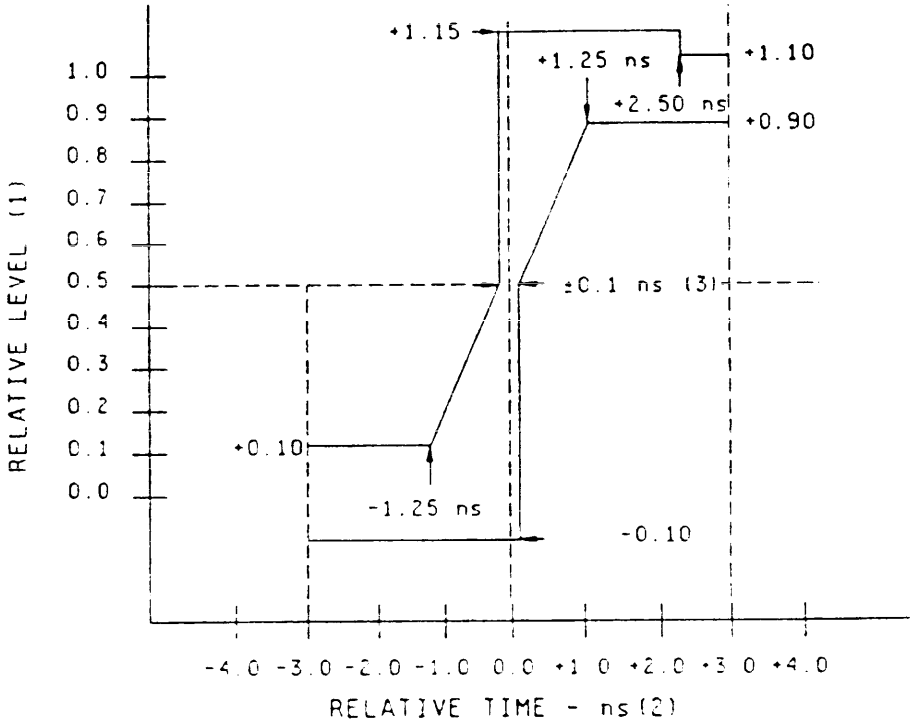
NOTES:

1. Normalized waveform magnitude, measured in terms of optical power. Derived from absolute waveform data as:

   $$\frac{\text{(Instantaneous magnitude) - (base magnitude)}}{\text{(top magnitude) - (base magnitude)}}$$

   where top and base magnitudes are as defined in notes 5 and 7.
2. Time measured relative to (center) pulse start/stop line.
3. These three lines are pulse start and stop lines (see note 9).
4. DATA input square wave frequency is 12.5 MHz ±50 ppm.
5. L(LR) interval for defining base magnitude at the 0.0 level.
6. L(L) interval during which optical power is at the lesser level.
7. L(HR) interval for defining top magnitude at the 1.0 level.
8. L(H) interval during which optical power is at the greater level.
9. Mesial line at 0.50 level.

FIGURE 1. Envelope waveform, top and base magnitude definition and templates.


+1.15
+1.25 ns
+1.10
+2.50 ns
+0.90
±0.1 ns (3)
+0.10
-1.25 ns
-0.10
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
RELATIVE LEVEL (1)
-4.0 -3.0 -2.0 -1.0 0.0 +1.0 +2.0 +3.0 +4.0
RELATIVE TIME - ns (2)

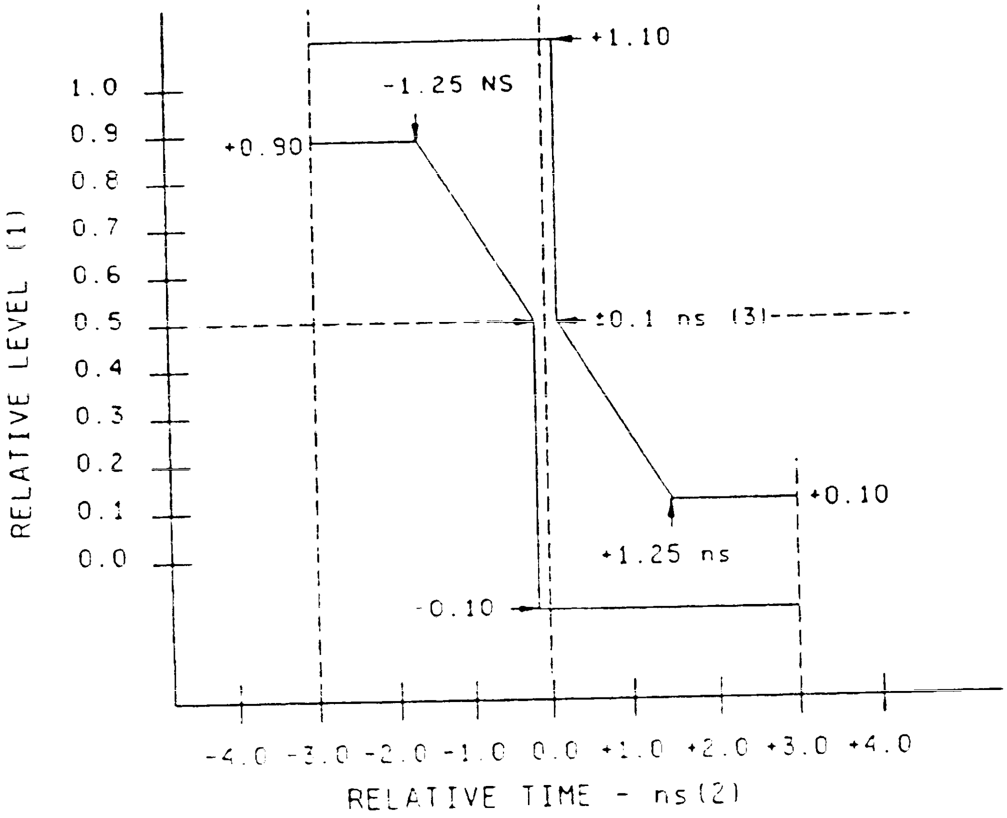
NOTES:

1. Normalized waveform magnitude, measured in terms of optical power. Derived from absolute waveform data as:

$$\frac{\text{(Instantaneous magnitude)} - \text{(base magnitude)}}{\text{(top magnitude)} - \text{(base magnitude)}}$$

where top and base magnitudes are as defined in figure 1.
2. Time is measured relative to (center) pulse start/stop line.
3. The ±0.1 ns at the 0.5 level is an accommodation for the width of the oscilloscope trace.

FIGURE 2. Positive transition waveform template.



NOTES:

1. Normalized waveform magnitude, measured in terms of optical power. Derived from absolute waveform data as:

$$\frac{(\text{Instantaneous magnitude}) - (\text{base magnitude})}{(\text{top magnitude}) - (\text{base magnitude})}$$

where top and base magnitudes are as defined in figure 1.

2. Time is measured relative to (center) pulse start/stop line.
3. The ±0.1 ns at the 0.5 level is an accommodation for the width of the oscilloscope trace.

FIGURE 3. Negative transition waveform template.

3.2.4.3 Pattern. DATA pattern modulation rate, duty factor and run length are specified in Table III.

TABLE III. Pattern characteristics.

Parameters	Min	Max	Units	Conditions
Modulation rate	125	160	MBd	see 6.3
Run length		10		see 6.3.5
Duty factor	0.35	0.65		see 6.3.1, averaged over 50 UI, minimum

3.2.5 Signal transfer. The relationship between the input signal and the output shall be as specified in the subparagraphs to this paragraph.

3.2.5.1 Data input to envelope waveform. The relationship between the envelope waveform and each of the data input waveforms shall be as specified in the subparagraphs to this paragraph.

3.2.5.1.1 Phasing. The phase of the envelope waveform with respect to the waveform at a DATA input shall be as specified in the subparagraphs to this paragraph.

3.2.5.1.1.1 Differential input. See table IV.

3.2.5.1.1.2 Single ended input. See table IV. The unused input is to be at the logic level identified in table IV as producing an envelope waveform top level.

TABLE IV. Phasing.

Data input 1/	logic level 1/	Envelope waveform 2/
+ DATA	- DATA	
H	L	T
L	H	B
H	H	B
L	L	B

1/ "H" and "L" indicate HIGH and LOW logic level, respectively.
2/ "T" and "B" indicate top and base level, respectively.

3.2.5.1.2 Optoelectronic propagation delay. The subparagraphs to this paragraph specify the optoelectronic delay propagation magnitude and difference.

3.2.5.1.2.1 Magnitude. Optoelectronic propagation delay (see 6.3) between transitions at the DATA inputs and corresponding transitions of the envelope waveform shall not be greater than 25 ns.

3.2.5.1.2.2 Difference. The difference between the optoelectronic delays associated with each input transition polarity shall not be greater than 0.6 ns when the DATA pattern is a square wave.

3.2.5.2 Disable input to optical output. The relationship between an applied signal at the transmit disable input and output power level shall be as specified in the subparagraphs to this paragraph.

3.2.5.2.1 Phasing. A logic HIGH at the transmit disable input shall disable the transmitter (see 3.2.3.3).

3.2.5.2.2 Response time. At 1 μs following a change in logic state of the input waveform, output power, averaged over a 40 ns interval, shall have completed no less than 90% of the change between the power levels corresponding to the logic states, as defined in table II. Reference levels for the transmit disable input are established at the crossing of the ECL level defined in MIL-M-24791, Appendix A.

3.2.6 Jitter (see 6.3).

3.2.6.1 Applicable conditions.

3.2.6.1.1 Reference waveform. Envelope waveform jitter shall be determined with respect to corresponding DATA input transitions.

3.2.6.1.2 Reference magnitudes The reference magnitude for the DATA input shall be that associated with the designated logic interface (see 3.2.2). The reference magnitude for the envelope waveform shall be its associated mesial level.

3.2.6.1.3 Frequency range. Unless otherwise specified, all jitter spectral components within the range extending from 10 Hz to a frequency equal to one half the modulation rate of the envelope waveform are to be included.

3.2.6.1.4 Transition durations. Requirements apply for all allowed transition durations.

3.2.6.2 Random (see 6.3.4) and systematic (see 6.3.6). The subparagraphs to this paragraph shall apply for a square wave pattern at any allowed modulation rate to all random and systematic jitter components.

3.2.6.2.1 Rms. The increase in jitter shall be no greater than 54.5 ps rms.

3.2.6.2.2 Spectral distribution Jitter spectral density with the range 500 Hz to 1.3 MHz shall be not greater than 4.4 X 10^5 fs^2/Hz.

3.2.6.3 Pattern dependent (see 6.3.3). The increase in pattern dependent jitter shall be no greater than 0.3 ns peak for an input signal pattern which is identified in annex a of ISO/IEC standard 9314-3.

3.2.6.4 Transfer to envelope waveform. The subparagraphs to this paragraph specify the relationship between certain specified parameters and jitter of the envelope waveform.

3.2.6.4.1 From data inputs. The magnitude of the transfer coefficient (see 6.3.2) relating jitter of the input to correlated jitter of the output shall be no greater than 1.1.

3.2.6.4.2 From power supply inputs. The magnitude of the transfer coefficient (see 6.3.2) between fluctuations of the voltage at any power supply terminal and the resulting jitter of the output signal shall be not greater than .5 ns/v for sinusoidal variations to 50 mv peak.

3.2.7 Power supply. The transmitter shall operate from a single supply.

3.2.7.1 Voltage. The functional requirements herein shall apply for power supply voltages from 4.75 Vdc to 5.25 Vdc inclusive at V_{CC} with respect to V_{EE}.

3.2.7.2 Current. The current demand shall be no greater than 250 mA dc.

3.2.7.3 Ripple and noise. The functional requirements apply for sinusoidal variations of the power supply voltage to 50 mV peak of either polarity for frequencies greater than 20 kHz.

3.2.8 Short circuit protection. Short circuit protection is not required for any terminal.

3.3 Environmental requirements. Environmental requirements of MIL-M-24791 shall apply as specified and as supplemented by the subparagraph to this paragraph.

3.3.1 Operating temperature range. Case operating temperature range 3 of MIL-M-24791 shall apply.

3.4 Design and construction. Design and construction requirements of MIL-M-24791 shall apply as specified and as modified or supplemented by the subparagraphs to this paragraph.

3.4.1 Package style. Package style AA shall be used as specified in MIL-M-24791, Appendix C.

3.4.2 Package isolation. There shall be no dc electrical connection between the package and any power supply terminal. Functional requirements herein shall be independent of connection of either of these terminals to the package.

3.4.3 Electrical terminals. Electrical connections shall be made through electrical terminals specified in this paragraph and its subparagraphs.

3.4.3.1 Assignment. Electrical terminal assignment shall be as specified in table V.

TABLE V. Terminal assignment.

TERMINAL NO.	DESCRIPTION	TERMINAL NO.	DESCRIPTION
1	V_{EE}	9	V_{EE}
2	DISABLE	10	V_{EE}
3	V_{EE}	11	V_{CC}
4	V_{EE}	12	V_{CC}
5	V_{EE}	13	V_{EE}
6	CASE	14	+DATA
7	NO PIN	15	-DATA
8	V_{EE}	16	V_{EE}

3.4.3.2 Type. The electrical terminal type shall be as specified for the package style in MIL-M-24791, Appendix C (see 3.4.1).

3.4.3.3 Finish. Electrical terminal finish shall be gold plate in accordance with MIL-H-38534.

3.5 Reliability. The mean time between failure (MTBF), as calculated using MIL-HDBK-217, shall be greater than 10^5 hours at 25 °C

3.6 Maintainability. The requirement stated in MIL-M-24791 applies.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. All quality assurance provisions of MIL-M-24791 apply as either modified or supplemented by the paragraphs contained within this section.

4.2 Quality assurance program. Quality assurance provisions of MIL-M-24791 shall apply as specified therein.

4.3 Inspection and controls during manufacture. Inspection and controls during manufacture provisions of MIL-M-24791 apply.

4.4 Device screening. Device screening of MIL-M-24791 shall apply except the stabilization bake is not required.

4.5 Quality conformance inspections (QCI). The subparagraphs to this paragraph supplement and modify the provisions of MIL-M-24791.

4.5.1 Group A inspection. The group A inspections specified in table VI shall be performed at the minimum, 5.0 V dc, and the maximum power supply voltages at the minimum, at +25°C, and at the maximum case operating temperature.

TABLE VI. Group A inspections.

Parameter	Requirement paragraph	Test paragraph
Optical spectrum	3.2.3.2	4.7.7.2
Power and extinction ratio	3.2.3.3	4.7.7.3
Envelope waveform	3.2.4	4.7.7.4
Optoelectronic propagation delay difference	3.2.5.1.2.2	4.7.7.7
Power supply current	3.2.7.2	4.7.7.14

4.5.1.1 Group A sample size and allowed failures. Table VII specifies samples sizes and the number of allowed failures per inspection lot for Group A inspection for a lot size of 50. Acquisition documentation must reference this table or provide an alternative.

Table VII. Group A sample sizes and allowed failures per inspection lot.

Parameter	Sample size	Allowed failures per lot
Optical spectrum	10	0 1/
Power and modulation extinction ratio	50	2/
Envelope waveform	50	2/
Optoelectronic propagation delay difference	10	0 1/
Power supply current	50	2/

NOTES:

1/ Group A QCI Inspection Failure of MIL-M-24791 applies except that failure of any device does affect the remainder of the inspection lot as with a Group B, C or D inspection. QCI resubmission of MIL-M-24791 applies.

2/ Group A QCI Inspection Failure of MIL-M-24791 applies.

4.5.2 Group B, C and D inspections. The Group B, C and D inspections of MIL-M-24791 shall apply.

4.6 Production representative device (PRD) screens and inspections. When specified in the acquisition documents (see 6.2), the inspection described in the subparagraphs to this paragraph shall be performed.

4.6.1 Screens. All device screens specified in MIL-M-24791, except the stabilization bake. The output power measurement of the optical power measurements method of MIL-M-24791, Appendix B, shall be performed before and after temperature cycling, constant acceleration and burn-in screens, and during the burn-in screen. Output power shall be as specified herein (see 3.2.3.3) The change in output power between before and after measurements shall be not greater than 0.5 dB.

4.6.2 Group A inspection. The Group A inspection tests (see 4.5.1) at minimum, nominal, and maximum power supply voltages at minimum, at 25 °C, and at maximum case operating temperature.

4.6.3 Group B inspection. The Group B inspection tests of MIL-M-24791 shall apply.

4.6.4 Design performance inspection. All tests in Table VIII shall apply.

TABLE VIII. PRD design performance inspection.

Parameter	Requirement paragraph	Test paragraph
Electrical Interface	3.2.2	4.7.7.1
Transfer - Phasing	3.2.5.1	4.7.7.5
Optoelectronic Propagation Delay - Magnitude	3.2.5.1.2.1	4.7.7.6
Transmit Disable - Response Time	3.2.5.2.2	4.7.7.8
Jitter - Rms Random and Systematic	3.2.6.2.1	4.7.7.9
Jitter - Random and Systematic Spectral Density	3.2.6.2.2	4.7.7.10
Jitter - Pattern Dependent	3.2.6.3	4.7.7.11
Jitter Transfer - From Data Inputs	3.2.6.4.1	4.7.7.12
Jitter Transfer - From Power Supply Inputs	3.2.6.4.2	4.7.7.13

4.7 Test conditions and methods. Test methods, and conditions applicable to them and those in MIL-M-24791 are specified in the subparagraphs to this paragraph.

4.7.1 Operational test conditions. Certain test methods contain the reference "Operational test conditions apply." If so, then the operating conditions and acceptance criteria of the subparagraphs to this paragraph apply.

4.7.1.1 Mounting. Certain test methods require that the method used to mount the device under test be identified. The device shall be mounted on a printed circuit board using either a socket or by soldering. Heat transfer shall be by natural convection; no auxiliary heat sinking shall be used. Sufficient clear space shall be provided around and above the device such that convection is not impaired and heat transfer between adjacent devices is insignificant.

4.7.1.2 Test circuit. The device shall be operated during testing as described in the optical power measurements test method of Appendix B of MIL-M-24791. Unless otherwise specified, the maximum power supply voltage shall be used.

4.7.1.3 Acceptance criteria. Optical output power (see 4.7.7.3) shall be measured, under the device operation provisions of MIL-M-24791, Appendix B, at the start and completion of the testing. The measured values shall lie within the limits of 3.2.3.3 and the magnitude of their difference shall be no greater than 0.5 dB.

4.7.2 Package integrity methods. All package integrity tests of MIL-M-24791 where indicated from the quality conformance inspections therein, apply without modification.

4.7.3 Stress test methods. All stress tests of MIL-M-24791 where indicated from the quality conformance inspections therein, apply. Supplemental information required by certain of these tests is specified in the subparagraphs to this paragraph.

4.7.3.1 Burn-in. Operational test conditions apply. The "test conditions" paragraph and its subparagraphs of MIL-STD-883, method 1015 do not apply. The test shall be performed at 85 °C +0/-2 °C for 240 hours. Post burn-in measurements shall be performed within 24 hours of completion of the test.

4.7.3.2 Temperature cycling. Operational test conditions apply except that device shall not be operated during the test. The following test temperature extremes shall be substituted for those given in table 1 of MIL-STD-883, method 1010: the hot shall be + 85 °C ± 5 °C, the cold shall be 0 °C ± 5 °C.

4.7.3.3 Steady state life. Operational test conditions apply. The "test duration", "test temperature" and "test conditions" paragraphs and their subparagraphs of MIL-STD-883, method 1005 do not apply. The test shall be performed at a minimum case temperature of 85 °C for 2000 hours.

4.7.4 Optical connection test method. Operational test conditions apply, except for the acceptance criteria. The pigtail cable/case junction strength test method of MIL-M-24791 shall be performed and its acceptance criteria applies. The parameter to monitored is optical power and shall be measured using the average power test procedure described in the optical power measurements test method of MIL-M-24791, Appendix B.

4.7.5 Chemical test methods. All quality conformance inspection chemical tests of MIL-M-24791 apply. The subparagraph to this paragraph provides supplemental information required by the moisture resistance test method.

4.7.5.1 Moisture resistance. Operational test condition apply. The minimum case temperature during step 7 of the test shall be 0 °C ± 2 °C.

4.7.6 Reliability test methods. The reliability test of MIL-M-24791 does not apply.

4.7.7 Functional test methods. The methods to be employed for evaluating functional performance are specified in the subparagraphs to this paragraph.

4.7.7.1 Electrical interface. Each test method listed in table IX shall be performed separately on the DATA+, DATA- and Transmit Disable inputs. Operational test conditions apply except for acceptance criteria. Acceptance criteria is stated in the conditions column of table IX.

TABLE IX. Electrical interface test methods.

Parameter	Test method	Conditions
Input Current, Logic HIGH	MIL-STD-883, Method 3010	Input voltage at input under test = V_{CC} - 0.75 Vdc; Input voltage at other inputs = V_{CC} - 1.38 Vdc; I_{IH}max = 400 µA
Input Current, Logic LOW	MIL-STD-883, Method 3009	Input voltage at input under test = V_{CC} - 1.8 Vdc Input voltage at other inputs = V_{CC} - 1.38 Vdc; I_{IL} max = 10 µA
Terminal Capacitance	MIL-STD-883, Method 3012	Bias level at input under test = V_{CC} - 1.38 Vdc; Input voltage at other inputs = V_{CC} - 1.38 Vdc; Maximum capacitance = 7 pF

4.7.7.2 Optical spectrum. Optical spectrum measurements shall be performed using test equipment having resolution and accuracy adequate to insure total measurement error magnitude does not exceed 10 percent. Central wavelength and spectral width shall be measured. Measurements of these parameters shall conform to their definitions contained in MIL-M-24791, Appendix B. Subject to procuring activity approval these measurements may be performed on the emitter prior to incorporation within the device package provided the device is subject to the same electrical and environmental conditions as it would experience in the assembled device.

4.7.7.3 Power and modulation extinction ratio. The average power measurement and extinction ratio measurement, as described in the optical power measurements test method of MIL-M-24791, Appendix B, shall be performed. Transmit disable function shall be verified as directed under the output enable control procedure of that test method.

4.7.7.4 Envelope waveform. All procedures of the modulation envelope test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.5 Transfer - phasing. The reconfiguration and phasing procedure for transmitters of the optoelectronic delay measurement test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.6 Data optoelectronic propagation delay magnitude. The transmitter delay procedure of the optoelectronic delay measurement test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.7 Data optoelectronic propagation delay difference. The data optoelectronic propagation delay difference shall be determined from the bias distortion procedure for transmitters of the modulation envelope waveform measurement test method of MIL-M-24791, Appendix B. The measured bias distortion is the data optoelectronic propagation delay.

4.7.7.8 Disable response time. The control parameter measurements test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.9 Rms random and systematic jitter. The random and systematic jitter measurements rms procedure for transmitters from the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed. The jitter measurement bandwidth shall be as specified in 3.2.6.1.3.

4.7.7.10 Random and systematic jitter spectral density. The random and systematic jitter measurements spectral density procedure for transmitters from the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed. The jitter measurement bandwidth shall be as specified in 3.2.6.2.2..

4.7.7.11 Pattern dependent jitter magnitude. The pattern dependent jitter magnitude measurement procedure for transmitters of the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.12 Jitter transfer coefficient from DATA input(s) to envelope waveform. The transfer measurements procedure for input to output for transmitters of the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.13 Jitter transfer coefficient from power supply input(s) to envelope waveform. The transfer measurements procedure for power supply to output for transmitters of the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.14 Power supply current. The power supply voltage and current test method for transmitters of MIL-M-24791, Appendix B, shall be performed. The input signal shall be a square wave at 125 MBd. Operational conditions apply (see 4.7.1). The power supply current shall be measured after the determination that output power is within the range specified in 3.2.3.3 for the applicable transmitter type.

5. PACKAGING

5.1 General. Packaging shall be in accordance with MIL-M-24791.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. The digital fiber optic transmitter described herein is intended for use in a SAFENET local area network application see MIL-STD-2204 for information on SAFENET). MIL-HDBK-818-1 provides network development and implementation guidance. They are also available for other applications which require digital fiber optic transmitters with nominal data rates not exceeding 160 MBd.

6.2 Acquisition requirements. The acquisition requirements of MIL-M-24791 shall apply.

6.3 Definitions. Terms used in this specification are defined in the subparagraphs to this paragraph and in MIL-M-24791 and the documents cited therein for this purpose. Order of precedence is this paragraph followed by the order given in MIL-M-24791.

Note: If a definition is not shown here, see paragraph 6.3 of MIL-M-24791.

6.3.1 Duty factor. The ratio of the sum of the durations of all pulses of the same polarity contained within a specified interval to the length of the interval (extended definition; see IEEE 194-1977).

6.3.2 Jitter transfer coefficient. The ratio between a specified component of output jitter and the stimulus that produces it, either jitter of the input signal or other specified parameter.

6.3.3 Pattern dependent jitter. That component of jitter that is correlated with the pattern of the stimulus signal.

6.3.4 Random jitter. That component of jitter that is uncorrelated with any component of the stimulus signal or device transfer parameter.

6.3.5 Run length. The number of unit intervals between transitions in a pulse train.

6.3.6 Systematic jitter. That component of jitter that is correlated with a device transfer parameter.

6.4 Abbreviations. Abbreviations for terms or phrases used in this specification are listed below or found in MIL-M-24791 or the documents cited therein for this purpose. Order of precedence is this paragraph followed by the order given in MIL-M-24791.

a. DODISS: Department of Defense Index of Specifications and Standards

b. LAN: Local area network

c. MTBF: Mean time between failures

d. QCI: Quality conformance inspection

e. SAFENET: Survivable Adaptable Fiber Optic Embedded Network

f. UI: Unit interval

6.5 Symbols. Symbols used in this specification are listed below or found in MIL-M-24791 or the documents cited therein for this purpose. Order of precedence in this paragraph followed by the order given in MIL-M-24791.

a. $\Delta\lambda$: Spectral width

b. λ_c: Central wavelength

6.6 PIN. The Part or Identifying Number (PIN) shall be in accordance with MIL-M-24791. Type LP transmitters shall have a -LP suffix added to the end of the part number; type HP shall have a -HP added to the end of the part number.

6.7 Approved source(s) of supply. The below approved source(s) of supply has submitted a quality assurance plan which has been reviewed and approved by the preparing activity, Naval Sea Systems Command.

Vendor cage number	Vendor name and address
0P926	Laser Diode, Incorporated Lightwave Products Division 4 Olsen Avenue Edison, NJ 08820

CONCLUDING MATERIAL

Custodian:
Navy - SH
Army - CR
Air Force - 85
NASA - NA

Preparing activity:
Navy - SH

Review activities:
Navy - EC, OS, YD

DLA - ES

Agent:
DLA - ES

(Project 6025-N003)

aluminum riveted joints and connections:

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. **DOCUMENT NUMBER** MIL-M-24791/1	2. **DOCUMENT DATE (YYMMDD)** 94/XX/XX

3. **DOCUMENT TITLE**
MODULE, FIBER OPTIC, TRANSMITTER, DIGITAL, 160 MBD, ASSOCIATED DETAIL SPECIFICATION FOR

4. **NATURE OF CHANGE** (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. **REASON FOR RECOMMENDATION**

6. **SUBMITTER**

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. **PREPARING ACTIVITY**

a. NAME Kendall A. Cottongim	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-5373 (2) DSN 986-5373
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Naval Sea Systems Command SEA 05Q42, Washington, DC 20362-5160 Telephone (703) 602-9137/6020 DSN 332-9137/6020

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